

When Federated Learning Meets Knowledge Distillation to Secure Consumer Edge Network

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Abstract—Consumer networks face several security challenges due to the distributed nature of edge devices and the sensitive data they handle. Federated Learning (FL) presents a promising paradigm for collaborative model training in distributed environments. However, its implementation in edge consumer networks raises concerns about model heterogeneity, communication efficiency, and reverse engineering attacks. To address these issues, in this paper, we introduce SKDFL, a novel framework that leverages Knowledge Distillation (KD) and Secure Multi-Party Computation (SMPC) techniques to enhance communication efficiency while preserving data privacy in edge consumer networks. Through the use of KD, the distilled knowledge is transmitted between devices, significantly reducing communication overhead. Additionally, we incorporate lightweight encryption mechanisms to protect soft-labels from reverse engineering attacks using SMPC. We evaluate our proposed framework using two public datasets and demonstrate its efficiency in reducing communication costs, achieving up to a 92.4% reduction compared to conventional FL methods. Moreover, SKDFL achieves high performances in terms of accuracy and F1-score in both binary and multi-class classification while preserving the privacy of clients. Our obtained results show the potential of SKDFL to address the challenges of communication efficiency and data privacy in FL for edge consumer networks, paving the way for secure and efficient collaborative learning in consumer networks.

Index Terms—TinyML, federated learning, knowledge distillation, edge consumer network.

I. INTRODUCTION

IN RECENT years, consumer networks have become more complex, connecting many different devices ranging from smartphones, smart home gadgets (such as thermostats and security cameras), and Internet of Things (IoT) devices. The rapid expansion of consumer networks has significantly improved connectivity and convenience. However, it also introduces critical security risks due to the growing number of connected devices. Many of these vulnerabilities stem

from outdated software, weak authentication systems, and inadequate security practices. As a result, attackers can exploit these weaknesses, leading to data breaches, identity theft, and unauthorized access to sensitive information. Addressing these challenges requires robust security measures to protect users and maintain network integrity [1], [2], [3].

The distributed nature of consumer networks, where devices frequently connect and disconnect from the network, complicates traditional security approaches. This dynamic environment makes it challenging for centralized security solutions to effectively monitor and defend against emerging threats in real-time. Additionally, the sheer volume and diversity of devices in consumer networks exacerbate the difficulty of enforcing consistent security policies across all endpoints. Consequently, gaps in security posture can emerge, leaving the network susceptible to infiltration and compromise. The interconnected nature of consumer networks introduces the risk of lateral movement, wherein attackers exploit vulnerabilities in one device to gain access to other interconnected devices or sensitive data within the network. This lateral movement can propagate rapidly, escalating the impact of a security breach and complicating remediation efforts. Additionally, the increasing prevalence of IoT devices in consumer networks introduces new attack vectors, as many of these devices lack robust security features and are often overlooked by users and manufacturers alike.

Addressing the security challenges inherent in consumer networks requires a multifaceted approach that combines proactive risk mitigation strategies with robust detection and response mechanisms. This approach can leverage advanced machine learning techniques, particularly for Intrusion Detection Systems (IDS), to enhance the network's defense capabilities. By deploying machine learning algorithms trained on historical data and real-time network traffic patterns, IDS can autonomously identify anomalous behavior indicative of potential security threats.

Machine learning-based IDS offers several advantages over traditional rule-based systems. They can adapt to evolving threats by continuously learning from new data, allowing them to detect previously unseen attack patterns and zero-day vulnerabilities. Additionally, machine learning algorithms can analyze vast amounts of network data at high speed, enabling rapid detection of security incidents without overwhelming network administrators with false positives. Moreover, machine learning techniques can be applied to anomaly detection, where deviations from normal network behavior are flagged as potential security threats. By establishing a

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baseline of typical network activity, machine learning algorithms can detect deviations that may indicate unauthorized access attempts, malware infections, or other suspicious activities. This proactive approach enables early detection and containment of security breaches, minimizing their impact on the network and reducing the time-to-response to security incidents [4], [5].

While the usage of ML models can indeed be effective in detecting attacks, it may not always be suitable for consumer devices with limited computing power, storage, and energy resources. Consumer devices, such as smartphones, IoT gadgets, and smart home appliances, often operate under stringent resource constraints, making it challenging to deploy and execute complex ML algorithms directly on these devices. One primary concern is the computational overhead associated with training and running ML models on resource-constrained devices. Training sophisticated ML models typically requires significant computational resources and large datasets, which may exceed the processing capabilities of consumer devices. Moreover, the memory and storage requirements for storing model parameters and intermediate data can strain the limited onboard resources of these devices, potentially impacting their performance and user experience. Additionally, energy consumption is a critical consideration for consumer devices, especially those powered by batteries. Running ML algorithms continuously on these devices can drain battery life rapidly, leading to decreased usability and user dissatisfaction. This energy consumption issue becomes even more pronounced in the context of real-time ML inference, where devices must process data streams continuously to detect and respond to security threats promptly.

To address the challenges of deploying ML models on resource-constrained consumer devices, researchers have proposed an innovative solution known as TinyML. TinyML focuses on optimizing ML algorithms and models for deployment on devices with limited computing power, storage, and energy resources, making it an ideal candidate for enhancing security on consumer devices. The key innovation behind TinyML lies in the development of lightweight ML algorithms and model architectures specifically designed to operate efficiently on microcontrollers and other embedded systems. These algorithms are optimized for low computational complexity, minimal memory footprint, and energy efficiency, enabling them to run effectively on devices with limited resources. By leveraging techniques such as quantization, pruning, and model compression, we can reduce the size and complexity of ML models without significantly sacrificing performance. On the other hand, Federated learning (FL) emerges as a key solution by enabling many edge devices to collaboratively learn a shared model while keeping the training data on local devices. This helps protect privacy and improve data security. Nevertheless, the vanilla FL approach still faces challenges such as high communication overhead and the heterogeneity of device capabilities, which can severely hamper the practical deployment and scalability of such models [6], [7], [8], [9].

To address these challenges, we propose SKDFL, a novel framework that integrates Knowledge Distillation (KD) into

the FL process to create a communication-efficient and secure learning environment for edge consumer networks. SKDFL is designed to enhance the efficiency of model training across diverse and resource-constrained edge devices without compromising the model's effectiveness or the privacy of the data. Through experimentation with the WSN-DS and UNSW-NB15 datasets, SKDFL demonstrates substantial reductions in communication costs, achieving up to a 92.4% reduction compared to conventional FL methods. Moreover, SKDFL maintains good model performance, thereby offering a promising solution for deploying privacy-preserving intelligent systems in edge environments. Knowledge Distillation serves as a strategic technique in this framework, enabling a "teacher-student" model paradigm where a compact student model learns from a more complex teacher model. By distilling knowledge from a comprehensive teacher network to lightweight student models, SKDFL significantly reduces the size and complexity of the models that need to be trained and updated across the network. This reduces the computing burden on each device and lowers the amount of data sent during training, helping to solve a major challenge in traditional federated learning.

The main contributions of this paper can be summarized as follows:

- We design SKDFL, a novel framework that leverages Knowledge Distillation (KD) techniques to enhance communication efficiency while preserving data privacy in edge consumer networks. Through the use of KD, the distilled knowledge is transmitted between devices, significantly reducing communication overhead.
- We incorporate lightweight encryption mechanisms to protect local model updates from reverse engineering attacks using Secure Multi-Party Computation (SMPC).
- We evaluate SKDFL using the WSN-DS and UNSW-NB15 datasets and demonstrate substantial reductions in communication costs, achieving up to a 92.4% reduction compared to conventional FL methods. Moreover, SKDFL achieves high model performances of 99% in accuracy and F1-score, offering a promising solution for deploying privacy-preserving and efficient intelligent systems in edge environments.

The structure of this paper is as follows: Section II reviews key studies that apply machine learning and TinyML for attack detection in consumer networks. Section III describes the proposed system model. Section IV evaluates the efficiency of SKDFL and compares it with recent approaches. Finally, Section V concludes the paper.

II. RELATED WORK

The rapid evolution of cyber-attacks necessitates advanced and dynamic defense mechanisms. Traditional IDS, which relied heavily on signature-based methods, struggled against new and evolving threats. Recent advancements in ML have significantly improved the capabilities of attack detection systems, making them more adaptive, efficient, and capable of handling vast and diverse data streams in real-time. This integration has facilitated the shift towards more robust anomaly detection frameworks that learn from data to identify

unusual patterns indicative of attacks. Deep Learning (DL), a subset of ML, has been pivotal in this transformation. Techniques such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs) have been employed to enhance the detection accuracy of complex attack patterns in network traffic. In [10], Jiang et al., showed that multi-channel LSTM networks can achieve high accuracy on the NSL-KDD dataset, a well-known benchmark in network security research. However, deep learning models often require significant computational power, making them difficult to use in resource-limited environments. This has led to growing interest in lightweight ML models that offer strong accuracy with lower computational requirements. The integration of machine learning into edge computing environments has rapidly evolved, driven by the increasing need for on-device intelligence and data privacy preservation. Traditional centralized training methods are often impractical due to their heavy reliance on data transmission and processing power, which are limited in edge computing scenarios. Federated learning, first introduced by McMahan et al. [11], offers a promising alternative by decentralizing the training process, thus allowing edge devices to collaboratively learn a shared prediction model while keeping all the training data on the device. FL's application in IDS is particularly valuable in environments where data privacy is paramount, such as in financial institutions or healthcare systems. Li et al. [12] explored FL's potential to perform secure, decentralized machine learning across multiple nodes, demonstrating that FL can effectively reduce the risk of data leakage during the training process. Moreover, FL facilitates the deployment of ML models in heterogeneous networks, where devices of varying computational capabilities collaboratively contribute to the model's training. Konečný et al. [13] introduced optimization techniques for FL that minimize the communication costs associated with model training, which is crucial for bandwidth-constrained environments [14], [15].

Despite its benefits, FL faces major challenges like high communication costs and differences in device capabilities. Konečný et al. [13] highlighted these challenges, noting that the frequent communication required to update model parameters across a distributed network often becomes a bottleneck, especially when dealing with large models or poor network conditions. To address these issues, researchers have explored various strategies to reduce the size of the transmitted data and to compress model updates effectively [12]. Parallel to advancements in federated learning, knowledge distillation has emerged as a robust technique for model compression and acceleration, where a compact "student" model is trained to emulate the behavior of a larger "teacher" model. Hinton et al. [16] demonstrated that knowledge distillation could effectively transfer the knowledge from a cumbersome model to a lighter model without significant loss in accuracy. This concept has been successfully applied across various domains, including natural language processing [17] and computer vision [16]. In the context of edge computing, the application of knowledge distillation presents a unique opportunity to enhance the efficiency of federated learning systems. The work by Li et al. [18] represents an early attempt

TABLE I
TABLE OF NOTATIONS

Parameter	Description
X	Input data
y	Ground truth label
$T(x)$	Teacher model (large neural network)
$S(x)$	Student model (small neural network)
θ_T	Parameters of the teacher model
θ_S	Parameters of the student model
$\hat{y}_T$	Prediction of the teacher model
$\hat{y}_S$	Prediction of the student model
KL	Kullback-Leibler Divergence
L_{KD}	Knowledge Distillation Loss
L_{CE}	Cross-Entropy Loss
L_{total}	Total Loss
T	Temperature parameter for knowledge distillation

to incorporate knowledge distillation within a federated learning framework, showing that it is possible to significantly reduce the model size and computational requirements on edge devices while maintaining high model performance. Our approach, SKDFL, not only minimizes the communication overhead associated with federated learning but also enhances the privacy and security of the local model updates, making it particularly suitable for secure edge consumer networks.

III. KNOWLEDGE DISTILLATION-ENABLED FEDERATED LEARNING IN EDGE CONSUMER

A. System Architecture

The SKDFL system is designed to facilitate a communication-efficient federated learning (FL) process across a network of edge devices, leveraging knowledge distillation techniques to enhance learning efficiency and model accuracy. This system model aims to address challenges associated with bandwidth constraints, limited computational resources, and the need for privacy preservation in consumer edge networks. We have considered a system architecture (see Fig. 1) in which only encrypted logits/soft-labels are exchanged between the server and clients. Thus, instead of communicating model parameters W_i at each round, we communicate knowledge in the form of soft-label outputs using a public shared dataset X^{pub} , as:

$$Y_i \leftarrow f_{W_i}(X^{pub}) \quad (1)$$

where f_{W_i} is the locally trained model parameterized by W_i .

B. System Model

Table I shows the table of notations. The SKDFL framework integrates the principles of federated learning (FL) with knowledge distillation (KD), allowing edge devices (nodes) to collaboratively learn a global model without sharing their local data. Each node trains a local model on its dataset and participates in a knowledge-sharing protocol that distills information from multiple models into a singular, more generalized model. This method effectively reduces the required communication bandwidth by transmitting compressed model updates (knowledge) instead of raw data, ensuring efficient use of network resources. In KD, initially, a global teacher

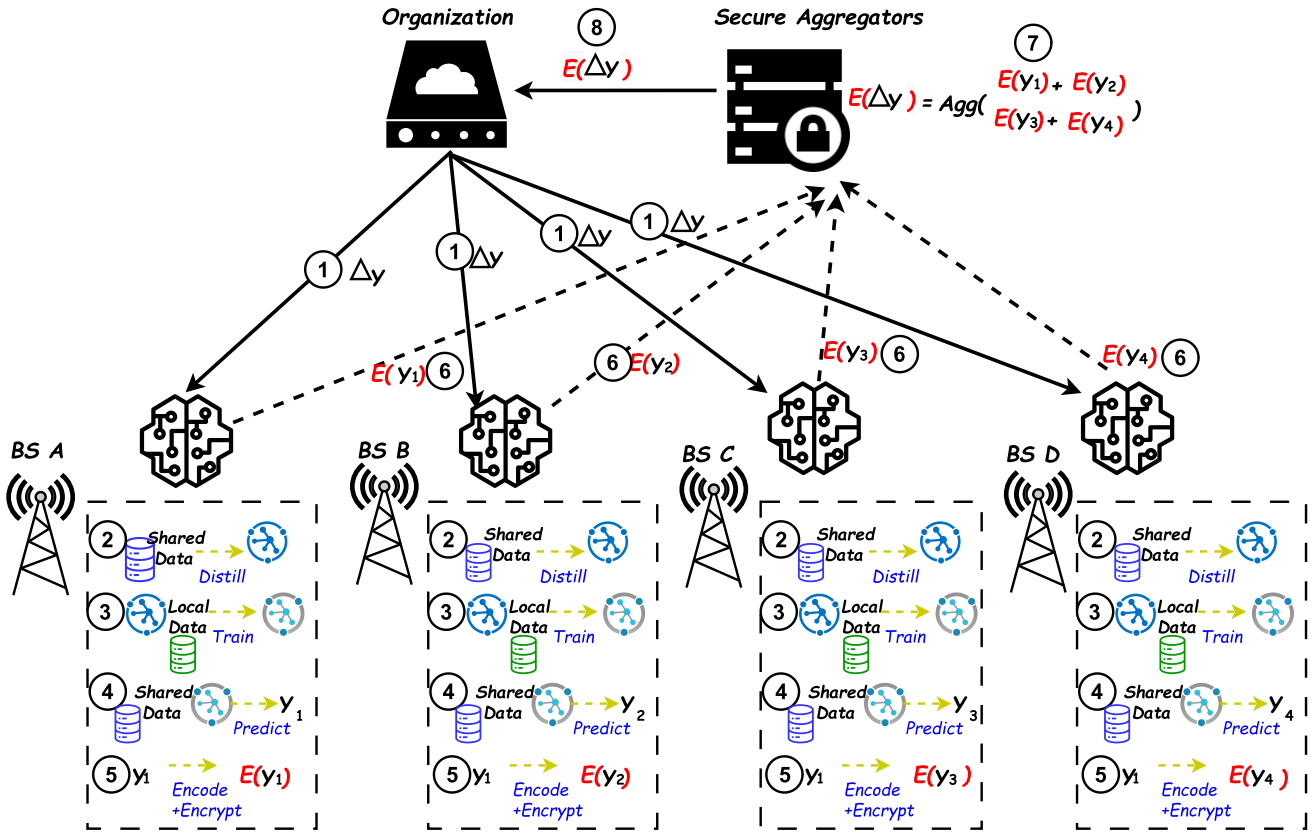


Fig. 1. SKDFL system architecture.

model is trained on a centralized server or cloud using a large dataset. This teacher model serves as the source of knowledge to be distilled to local student models. Each node in the network trains its student model using its local dataset. The loss function optimized during this training is given by the cross-entropy, which for a binary classification task can be expressed as:

$$\mathcal{L}_{local}(\theta_i) = - \sum_{(x,y) \in D_i} [y \log f_i(x; \theta_i) + (1 - y) \log (1 - f_i(x; \theta_i))]; \quad (2)$$

where $f_i(x; \theta_i)$ is the sigmoid output of the local model for input x , θ_i are the parameters of the model at node i , and (x, y) represents the training pairs in the local dataset D_i .

After local training, each device conducts knowledge distillation by leveraging the pre-trained global teacher model. The student model on each device learns to mimic the behavior of the global teacher model by minimizing a distillation loss function. This distillation process typically involves comparing the output distributions or representations of the teacher and student models and optimizing the student model's parameters to match those of the teacher. Knowledge distillation allows student models to learn from the output distributions (soft targets) generated by teacher models. These distributions are softened using a temperature parameter T , enhancing the information transfer about inter-class relationships:

$$p_T(x) = \sigma\left(\frac{z_T(x)}{T}\right), \quad q_S(x) = \sigma(z_S(x)), \quad (3)$$

where σ is the softmax function, $z_T(x)$ and $z_S(x)$ are the logits from the teacher and student models, respectively.

The knowledge distillation loss is then computed as:

$$\mathcal{L}_{KD} = T^2 \cdot \text{KL}\left(\sigma\left(\frac{z_T(x)}{T}\right) \parallel \sigma(z_S(x))\right), \quad (4)$$

combined with the traditional hard target cross-entropy loss:

$$\mathcal{L}_{combined} = \alpha \cdot \mathcal{L}_{hard}(q_S, y) + \beta \cdot \mathcal{L}_{KD}. \quad (5)$$

Once knowledge distillation is complete on each device, the updated student models are aggregated to obtain a new global model. This aggregation process may involve averaging the model parameters or using more sophisticated techniques to combine the models effectively. Aggregation can be performed in a decentralized manner, with devices communicating directly with each other or through a central server acting as an aggregator. After local updates, the models are aggregated securely to update the global model. To ensure privacy, the updates are encrypted before transmission and decryption is performed during aggregation:

$$\Delta\theta_g = \frac{1}{N} \sum_{i=1}^N \text{Enc}(\Delta\theta_i), \quad (6)$$

$$\theta_g \leftarrow \theta_g + \eta \cdot \text{Dec}(\Delta\theta_g), \quad (7)$$

where Enc and Dec represent the encryption and decryption functions, and Δ is the learning rate for updating the global model.

In the SKDFL framework, secure aggregation of model updates is crucial to maintain the privacy and security of the data. We utilize Secure Multi-Party Computation (SMPC) to perform this aggregation without revealing individual updates to any of the participating nodes or to an external adversary. The security of this SMPC protocol relies on the assumption that a sufficient number of nodes behave honestly and do not collude. If a majority of the nodes are honest, they can prevent any malicious node from learning about individual updates or from influencing the aggregated result unduly.

Assuming each node i has a model update $\Delta\theta_i$, the goal is to compute the sum of these Soft Labels securely. The process involves the following steps:

- 1) Each node i splits its update $\Delta\theta_i$ into N shares such that the sum of these shares equals the update:

$$\Delta\theta_i = \sum_{j=1}^n s_{ij}, \quad s_{ij} \in \mathbb{R}^d, \quad (8)$$

where s_{ij} is the share of node i 's update given to node j , and d is the dimensionality of the model parameters.

- 2) Each node i securely sends the share s_{ij} to node j .
- 3) Each node j aggregates the shares it receives from all nodes:

$$S_j = \sum_{i=1}^n s_{ij}. \quad (9)$$

- 4) The nodes then combine their aggregates to compute the overall update:

$$\Delta\theta_g = \sum_{j=1}^n S_j = \sum_{j=1}^n \sum_{i=1}^n s_{ij} = \sum_{i=1}^n \Delta\theta_i. \quad (10)$$

This method ensures that the individual updates $\Delta\theta_i$ are never revealed to other nodes or any external observer. The use of SMPC allows the computation of the sum of updates in a way that preserves the confidentiality of the data. Algorithm 1 shows the main training process of our SKDFL.

IV. IMPLEMENTATION

We have implemented our proposed SKDFL, using PySyft, a privacy-centric deep-learning library built on top of the PyTorch framework. In our test environment, we considered two datasets: the WSN-DS dataset and the UNSW-NB15 dataset. The WSN-DS dataset is used as the public dataset shared between the organization and clients, while the UNSW-NB15 dataset is treated as the local dataset. We divided the UNSW-NB15 dataset among clients so that each client has a subset containing samples of each attack type. To evaluate SKDFL, we defined two scenarios: binary and multi-class classification. In each scenario, we varied the number of involved FL clients from 5 to 10 and adjusted the rounds of training from 10 to 50 (see Table II). During each round, clients distill knowledge using the shared WSN-DS dataset, apply their subset of the local UNSW-NB15 data, and then use the WSN-DS dataset to predict new labels. Figs. 2 and

Algorithm 1: SKDFL Pseudo-Algorithm

Input: Set of node models $\{M_i\}_{i=1}^N$, local datasets $\{D_i\}_{i=1}^N$.

Output: Optimized federated model parameters with distilled knowledge

Initialize: Set a public dataset X^{pub} .

- 1 **Initial Phase:** Generate a fixed finite field (*i.e.*, a set of integers from 0 to $Q - 1$ for a prime number Q) for all computations to take place along with the initial global model parameters W_0 .
- 2 **for** $r \leftarrow 1$ **to** $r_{\max}$ **do**
- 3 **for each** Secure Aggregator n in N **do**
- 4 **for each** Consumer Edge Node i in I **do**
- 5 $Y_{r,i,n}^{pub,c} \leftarrow \text{NodeUpdate}(r)$;
- 6 $Y_{r+1,n}^{pub,c} = \frac{1}{I} \sum_{i=1}^I Y_{r,i,n}^{pub,c}$;
- 7 $Y_{r+1}^{pub,c} = \sum_{n=1}^N Y_{r+1,n}^{pub,c}$;
- 8 $Y_{r+1}^{pub} \leftarrow D(\text{Decode}(Y_{r+1}^{pub,c}))$;
- 9 Send Y_{r+1}^{pub} to each Consumer Edge Node;
- 10 **ClientUpdate (r):** // Each Consumer Edge Node (Student) executes this function;
- 11 **for each** Local client i in I in parallel: **do**
- 12 1. Perform knowledge distillation from the public shared dataset: $W \leftarrow \text{train}(W, X_r^{pub}, Y_r^{pub})$;
- 13 2. Local Training: $W_{r,i} \leftarrow \text{train}(W, X_{r,i}, Y_{r,i})$;
- 14 3. Compute Soft Labels: $Y_{r,i}^{pub} \leftarrow f_{W_i}(X_{r,i}^{pub})$;
- 15 4. Encode Soft Labels: $Y_{r,i}^{pub,c} \leftarrow \text{Encode}(Y_{r,i}^{pub})$;
- 16 5. Encrypt Soft Labels: Split $Y_{r,i}^{pub,c}$ into N shares;
- 17 6. return $Y_{r,i,n}^{pub,c}$;

TABLE II
CONSIDERED SCENARIOS

Scenarios	Type	FL clients	Rounds
1	Binary	5-10	10-50
2	Multi-class	5-10	10-50

3 show the considered student model (*i.e.*, local model) training loss for 5 and 10 clients, respectively, during 10 rounds, 25 rounds, and 50 rounds, respectively. We observe that the student model loss, in both cases, decreases until it reaches the minimum only after a few rounds of training (*i.e.*, 2 rounds). This demonstrates the communication efficiency of the SKDFL framework, as it achieves significant loss reduction with minimal communication rounds. Such efficiency is crucial in FL, where reducing the number of communication rounds between clients and the central server helps minimize bandwidth usage and latency, leading to more scalable and practical FL implementations.

A. Performance Evaluation

The performance of our proposed framework is assessed using metrics like accuracy, precision, recall (true positive rate, TPR), F1-score, and the area under the receiver operating

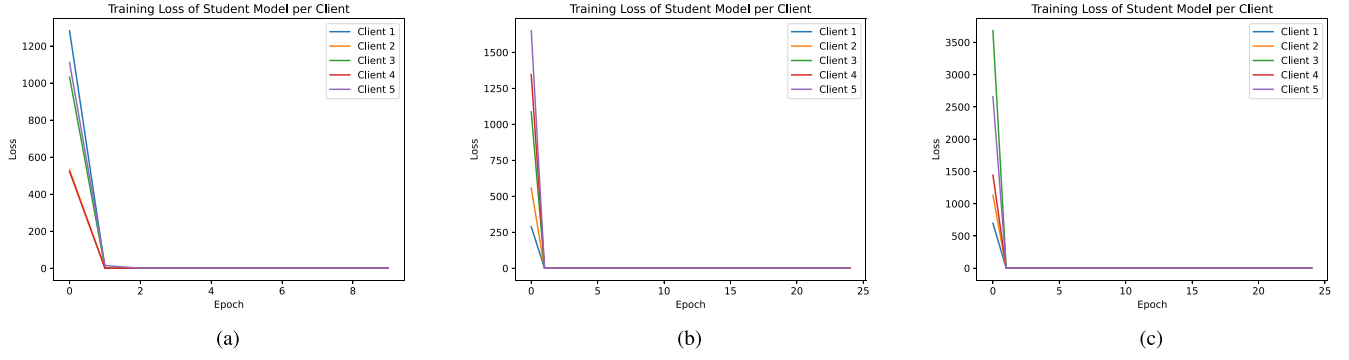


Fig. 2. Training loss of local model for 5 clients during (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

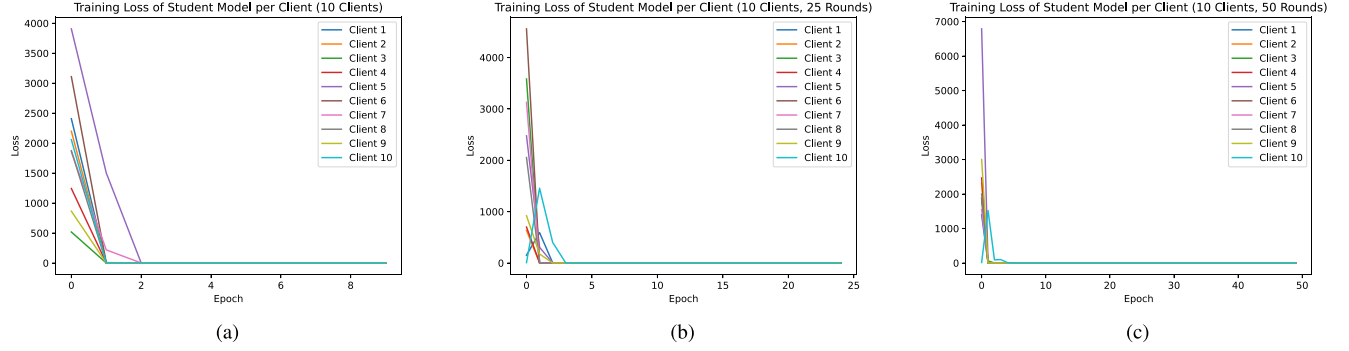


Fig. 3. Training loss of local model for 10 clients during (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

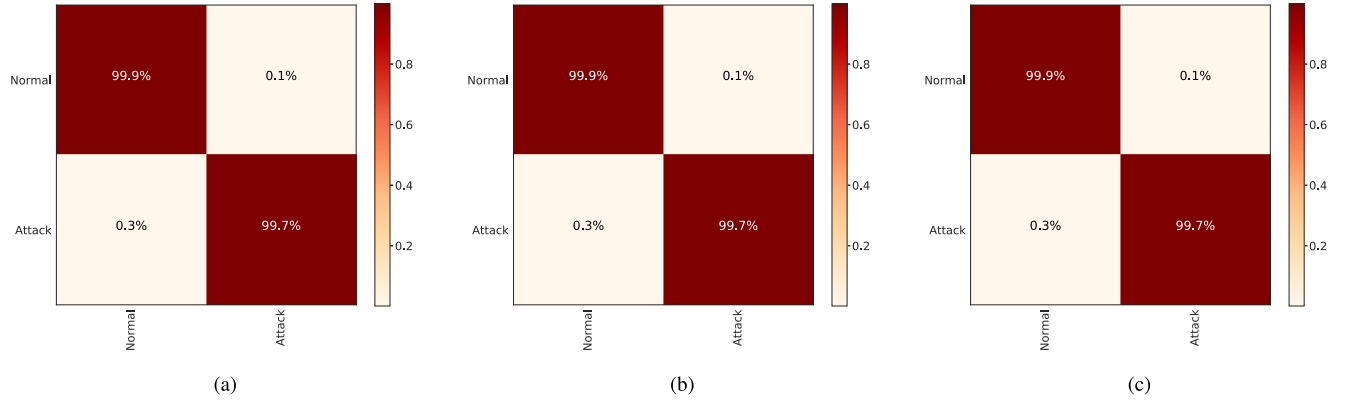


Fig. 4. Confusion matrices of our SKDFL for (a) 10 training rounds; (b) 25 training rounds; and (c) 50 training rounds.

characteristic (ROC) curve (AUC). The ROC curve visually plots the true positive rate against the false positive rate (FPR). A confusion matrix is also used, providing counts of true positives, false positives, true negatives, and false negatives.

Accuracy measures the model's ability to classify data correctly and is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

Precision measures the proportion of correct positive classifications:

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

Recall (or TPR) indicates the proportion of actual positives correctly identified:

$$Recall = DR = TPR = \frac{TP}{TP + FN} \quad (13)$$

The F1-score, the harmonic mean of precision and recall, balances both metrics:

$$F1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} \quad (14)$$

where True positives (TP) are correctly identified intrusions, false negatives (FN) are missed intrusions, false positives (FP) are incorrect intrusion classifications, and true negatives (TN) are correctly classified benign instances. A high-performing model will achieve strong accuracy, recall, and F1-score while keeping false positives low. Fig. 4 shows the confusion

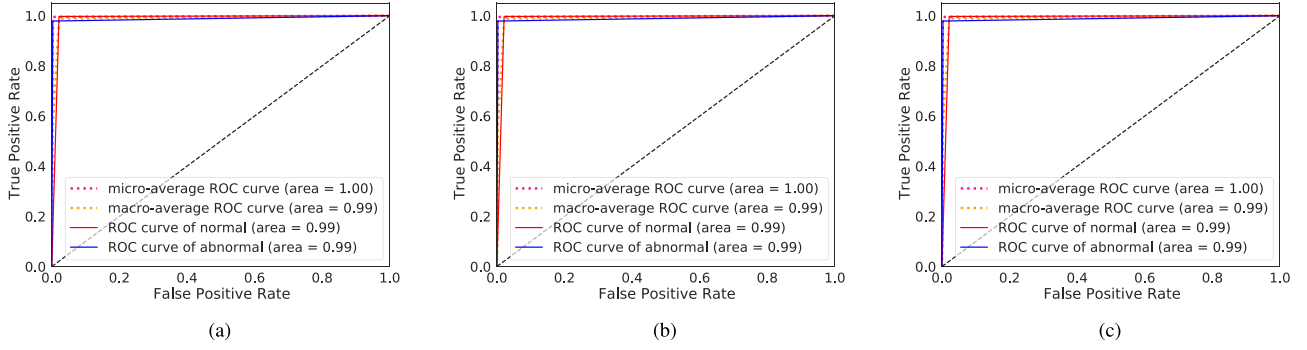


Fig. 5. ROC Curves of our SKDFL for (a) 10 training rounds; (b) 25 training rounds; and (c) 50 training rounds.

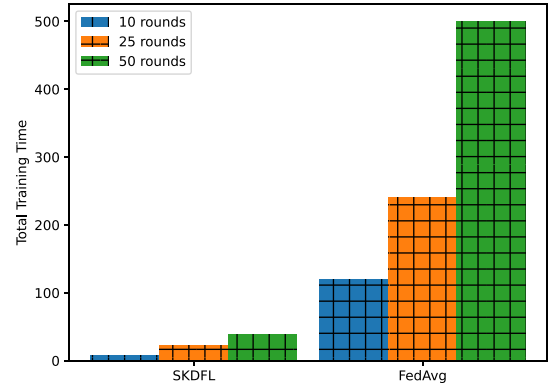
TABLE III
PERFORMANCE METRICS OF SKDFL AND RECENT AI MODELS
USING THE UNSW-NB15 DATASET

Methods	Accuracy	TPR	Time (second)
Fuzziness Semi-Supervised Learning [19]	0.86	0.85	N/A
Random Forest Based IDS [20]	0.93	0.92	N/A
IDSWSN [21]	0.92	0.91	N/A
IDS Outlier Gaussian Mixture [22]	0.95	0.94	N/A
IDSCPS [23]	0.96	0.95	N/A
SKDFL	0.99	0.99	28.08

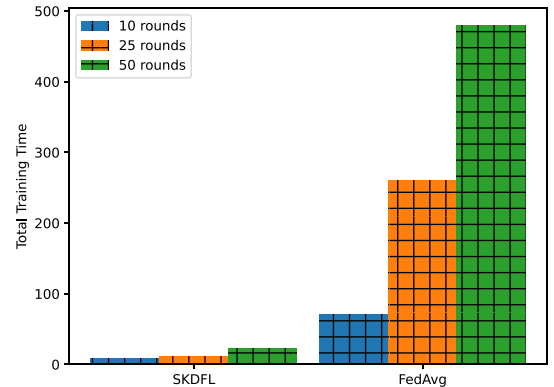
matrices of our SKDFL framework across different training rounds: (a) 10 training rounds, (b) 25 training rounds, and (c) 50 training rounds. SKDFL achieves high accuracy and an F1-score of 99%. Fig. 5 presents the ROC curves of our SKDFL framework across different training rounds: (a) 10 rounds, (b) 25 rounds, and (c) 50 rounds. The curves illustrate the model's performance in terms of the true positive rate and the false positive rate for various classification thresholds. A high-performance score of 99% was achieved across all three training durations. Macro-average ROC curves were also generated by calculating the TPR and FPR for each class and averaging them, which is particularly useful for balanced classes and assessing the classifier's overall performance. We consistently achieved a high performance score of 99% for 10, 25, and 50 rounds. Figs. 6(a) and 6(b) show the total time taken to complete the federated learning (FL) process for our SKDFL framework compared to traditional FL (specifically FedAvg (without KD)) for both binary and multi-class classification tasks. The results demonstrate significant reductions in communication costs, achieving up to a 92.4% reduction compared to conventional FL methods.

B. Comparative Analysis

In this section, we compare the performance of our proposed SKDFL framework with several centralized AI-based schemes in detecting attacks in consumer networks. We



(a)



(b)

Fig. 6. Total duration of SKDFL process training and FedAvg in the (a) binary classification and (b) multi-class classification.

have compared our proposed framework (SKDFL) with recent AI-based intrusion detection approaches from the literature, including Fuzziness Semi-Supervised Learning [19], Random Forest-Based IDS [20], IDS for Wireless Sensor Networks (IDSWSN) [21], IDS using Outlier Gaussian Mixture [22], and IDS for Cyber-Physical Systems (IDSCPS) [23]. Table III presents the accuracy and F1-score for our SKDFL framework compared to the centralized AI approaches on the UNSW-NB15 Dataset. The results demonstrate that our framework

achieves the highest accuracy of 99% and the highest F1-score of 99% with only 28.08 seconds of training time. This shows that our proposed SKDFL framework not only maintains the privacy of participants but also outperforms centralized methods in terms of detection accuracy and F1-score. Overall, SKDFL is a promising framework, offering robust protection against emerging attacks while preserving data privacy.

V. CONCLUSION

This paper proposed a novel federated learning approach for secure edge consumer network, called SKDFL. The SKDFL framework uses a novel integration of federated learning with knowledge distillation tailored to the needs of edge computing in consumer networks. This approach not only addresses privacy and communication efficiency but also leverages advanced machine-learning techniques to cope with the computational limitations of edge devices. For future work, we intend to explore optimizing the trade-offs between model accuracy and computational efficiency to further enhance performance on resource-constrained edge devices. Additionally, we intend to integrate Explainable AI techniques, which could improve transparency and trust in the model's decisions, addressing concerns about the interpretability of local models at the edge.

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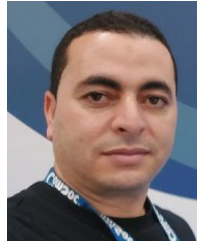


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